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Question bank

CSA3007 Deep learning

<table>
<tr><th>UNIT</th><th>Question Description</th></tr>
<tr><td>1</td><td>Explain the concept of eigenvalues and eigenvectors and discuss their importance in Deep Learning. Also, find the eigenvalues and corresponding eigenvectors of the following matrix:</td></tr>
<tr><td></td><td>Perform the following steps of PCA for the given dataset X:

• Mean centring
• Covariance matrix
• Eigenvalues & eigenvectors

X=[[2, 0] , [0, 2], [3, 1]].</td></tr>
<tr><td></td><td>The probability mass function of a discrete R.V X is given in the following table
<table>
<tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr>
<tr><td>P(X)</td><td>a</td><td>3a</td><td>5a</td><td>7a</td><td>9a</td><td>11a</td><td>13a</td><td>15a</td><td>17a</td></tr>
</table>
Find (i) the value of a ,(ii) $P(X < 3)$ (iii) Mean of X, (iv) Variance of X.</td></tr>
<tr><td></td><td>Derive the mathematical formulation of convolution in CNNs and analyze how filters, stride, pooling, dropout, and deep layered architectures influence feature hierarchy and generalization. Justify the suitability of CNNs over fully connected ANNs for high-dimensional image data with respect to parameter efficiency and spatial invariance.</td></tr>
<tr><td></td><td>Given the matrix A = [[3, 1], [0, 2]], apply eigen decomposition to compute the eigenvalues of matrix A and determine the corresponding eigenvectors. Further, define eigenvalues and eigenvectors, explain the concept of eigen decomposition, and state the necessary condition under which a matrix can be diagonalized.</td></tr>
<tr><td></td><td>Explain the concept of eigenvalues and eigenvectors and discuss their importance in Deep Learning. Also, find the eigenvalues and corresponding eigenvectors of the following matrix:</td></tr>
<tr><td></td><td>Critically compare supervised and unsupervised learning paradigms with respect to hypothesis space, capacity, and generalization bounds. Explain Bayesian classification from a probabilistic perspective and outline the systematic steps involved in building a robust machine learning algorithm, including model selection and validation strategies.</td></tr>
<tr><td></td><td>Find the eigenvalues and eigenvectors of Matrix A and B:
A = [[2, 1], [1, 2]]
B= [[1, 0, 0], [0, 5, 0], [0, 0, 9]]</td></tr>
</table>

	Exemplify the concept of overfitting and underfitting with the help of graphs, bias, variance as well as draw the bias and variance trade off curve.
	Consider a machine learning model trained on a dataset for predicting house prices. After training, the following results are obtained: Model 1: • Training Error = 4% • Validation Error = 22% Model 2 (Simpler Model): • Training Error = 18% • Validation Error = 20% 1. Identify which model suffers from overfitting and which suffers from underfitting**.** Justify your answer clearly. 2. Explain the bias–variance trade-off in the context of overfitting and underfitting. 3. Suggest at least three techniques to reduce overfitting in deep learning models. Explain briefly how each technique helps. How does increasing model complexity affect training error and validation error? Explain your answer with reasoning.
	Compare and contrast over fitting with under fitting from the following aspects. a. Definition b. Identifying each of them c. Common causes d. Training error, testing error, model complexity e. Fixing each of the error.
	Differentiate between Batch Gradient Descent, Stochastic Gradient Descent (SGD) and Mini-batch Gradient Descent.
	Consider the cost function used in linear regression: *J(w) = (w - 3)^2* Let the initial weight be w = 6. Using Gradient Descent, perform three iterations of the update rule for each of the following learning rates: $\eta = 0.1, \eta = 0.5, \text{ and } \eta = 1.2$ For each learning rate, compute and find the updated weights w, w, and w. Based on the obtained values of w, w, and w, compare the convergence behavior for different learning rates and explain what happens when the learning rate is too small or too large.
2	Develop a CNN for classifying MNIST dataset handwritten images. Explain the functionalities of a convolution, max pooling, flatten and dense layers.
	Exemplify the following. A. Structure of a biological neuron and its relation to ANN. B. Principle components of ANN. C. Difference between single-layer and multi-layer neural networks. D. Vanishing gradient problem in Sigmoid E. Graphs and Mathematical Equations of below mentioned Activation Functions • Sigmoid • Tanh • ReLU • Leaky ReLU • Softmax
	Explain the concept of input gate, forget get and output gate in LSTM along with the mathematical equations.

Explain the unfolding of a Recurrent Neural Network (RNN) across time steps and derive the Backpropagation Through Time (BPTT) algorithm for sequence learning. Critically analyse the vanishing and exploding gradient problems and justify how LSTM architecture overcomes these limitations using gating mechanisms.

Consider a training dataset representing student performance as shown below:

1	0	-1
1	0	-1
1	0	-1

Hours of Study (x)	**Hours of Sleep (x)**	**Desired Output (d)**
4	6	0
3	4	1
7	6	1
6	7	1

With initial weights bias, and learning rate $\eta = 0.6$. Use a step activation function to perform one epoch of training using the Perceptron Learning Algorithm. Show all intermediate calculations and the updated values of the weights and bias after processing each training example.

Explain the architecture of a Long Short-Term Memory (LSTM) model. Describe the role of the input gate, forget gate, and output gate in controlling the flow of information. Also, derive the mathematical equations governing the cell state update and explain how LSTM overcomes the vanishing gradient problem in traditional RNNs.

Explain the Convolutional Neural Network (CNN) with the help of following terminologies.

A. Why are CNNs preferred over traditional ANN for image data?
B. What are the main layers in CNN?
C. What is convolution operation?
D. What is a feature map?

Explain why LSTM was introduced to overcome the limitations of traditional Recurrent Neural Networks (RNNs). Describe the main components of LSTM, including the forget gate, input gate, and output gate, and briefly mention its applications in machine learning.

Consider the following 4×4 input image:

1	2	0	1
3	1	2	2
0	1	3	1
2	2	1	0

and a 3×3 filter (kernel):

Assume: Stride = 1 and Padding = 0 (Valid convolution)

a) Compute the output feature map after performing the convolution operation.
b) Determine the size (dimension) of the output feature map.
c) Apply the ReLU activation function on the obtained feature map.
d) Apply 2×2 Max Pooling with stride = 1 on the ReLU output and determine: The pooled feature map and size (dimension) after pooling.

Explain how convolution, activation, and pooling together enable hierarchical feature extraction in CNN.

Explain the concepts of RNN and vanishing gradient problem with the help of example.

Explain the architecture of an LSTM network and it's functionality in detail.

3	Explain the mathematical background of the variational auto encoder with the functions of the encoder, decoder, latent space and loss function.
	"A Variational Autoencoder (VAE) is a generative deep learning model that learns to map input data to a continuous, probabilistic latent space, allowing it to compress data and generate new, similar samples." Justify the statement with describing its working principle and the roles of the encoder, decoder, and latent space representation in VAE.
	Analyse the architecture and training mechanism of a Deep Autoencoder (DAE) for nonlinear feature extraction and dimensionality reduction, and critically compare it with VAE in terms of latent space structure, reconstruction loss, generative capability, and generalization behaviour.
	Explain main components of Variational Autoencoder (VAE), Latent Space, Reconstruction Loss and Activation Function.
	Explain the concept of a Variational Autoencoder (VAE) as a generative model. In your answer, discuss: 1. The difference between a traditional autoencoder and a variational autoencoder. 2. The VAE loss function, including the reconstruction loss and KL-divergence term. 3. Applications of VAEs in generative modelling and representation learning.